MILITARY SPECIFICATION

FORGINGS, PREALLOYED STEEL POWDER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers forged metal powder parts fabricated from 4600 (Modified) prealloyed powder plus blended carbon.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal

Fed. Test Method Std. No. 151 - Metals; Test Methods

American Society for Testing and Materials

ASTM Standard Test Method B-328

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition. The forged metal powder parts, and the mechanical test specimens, shall conform to the requirements of Table I. The parts and test specimens shall be fabricated from 4600 (Modified) prealloyed powder plus blended carbon.

Table I
Chemical Composition

Element	Analysis (Percent)
Carbon	0.38-0.43
Nickel	1.65-2.00
Molybdenum	0.4 -0.6
Manganese	0.15-0.30
Phosphorus	0.04 (max.)
Sulfur	0.04 (max.)
Oxygen	0.030 (max.)
Silicon	0.35 (max.)

FSC FORG

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3.2 Mechanical properties. The mechanical properties, at room temperature, of the austenitized, quenched, and tempered test specimens representing the metal powder parts shall conform to Table II.

Table II
Mechanical Properties
(Minimum)

Yield Strength[1] (ksi)	Tensile Strength (ksi)	Elongation[2] (Percent)	Reduction of Area (Percent)	Impact Strength[3,4] (Ft. lbs.)	Hardness[5] (Rockwell-C)
110	130	10.0	35.0	25 @ +72°F 8 @ -70°F	30-33

1 0.2% offset
2 2" gage length
3 Charpy V-notch
4 Specified temperatures have a tolerance range of ±2°F
5 Although the drawing or requirement for a particular part may specify different hardness levels, the test specimens, and only the test specimens, shall be heat treated to the hardness specified in this table.

3.3 Density. The density of the parts and the test specimens shall be at least 7.81 g/cc. The density of sample specimens taken from the part shall be at least 7.80 g/cc.

3.4 Decarburization. Unless otherwise specified, surface decarburization, total and partial, of the parts and the test specimens shall not exceed 0.001 inch.

3.5 Microstructure. The microstructure of the quenched and tempered parts and test specimens shall be the same and exhibit homogeneity equal to or better than Figure 1. Figure 2 is a representative example of unacceptable microstructure.

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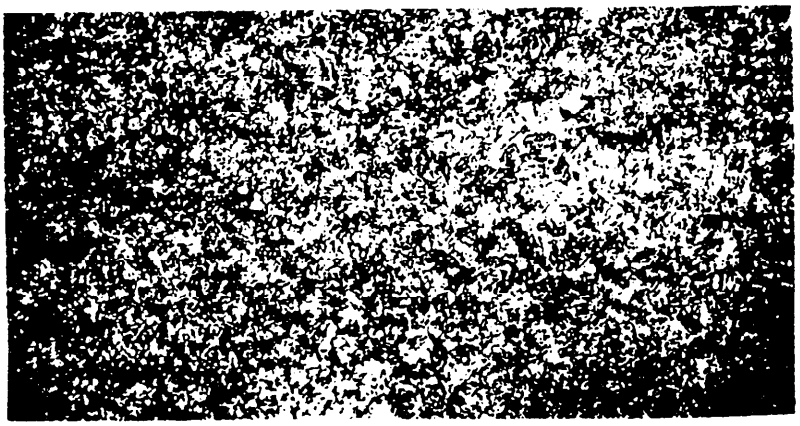


FIGURE 1 ETCHANT: 2% Nital MAG. 100X



FIGURE 2 ETCHANT: 2% Nital MAG. 100X

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection provisions.

4.2.1 Inspection lot. Unless otherwise specified, a lot shall consist of that quantity of finished (complete) parts produced from one batch of blended powder.

4.2.2 Sampling.

4.2.2.1 For chemical analysis. At least two samples for chemical analysis, consisting of one part and one mechanical test specimen, shall be selected from each lot, after final heat treating procedures. The sampling procedure shall be in accordance with Method 111 or 112 of Federal Test Method Standard No. 151.

4.2.2.2 For mechanical properties. A quantity of test specimens numbering at least 6 per each 100 production parts, or other quantity as mutually agreed upon by the manufacturer and purchaser, consisting of 3 for each of the tensile and impact tests, shall be prepared and processed concurrently with, and in an identical manner except for final tempering temperature, the production parts.

4.2.2.3 For density. Each production part and each test specimen shall be tested for density in accordance with 4.3.3. Test samples taken from representative locations in the part shall be selected as mutually agreed upon by the manufacturer and purchaser.

4.2.2.4 For surface decarburization. At least one test sample from each heat treat batch shall be tested in accordance with 4.3.4.

4.2.2.5 For microstructure. Each Charpy V - notch impact specimen, and one production part from each corresponding 100 parts, shall be prepared and examined in accordance with 4.3.5.

4.3 Tests.

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4.3.1 Chemical analysis. The samples shall be prepared and tested as specified in Method 111 or 112 of Federal Test Method Std. No. 151 for all elements except oxygen. In case of dispute, analysis by Method 111 shall be the basis for acceptance or rejection. Vacuum fusion analysis or other suitable methods as mutually agreed upon by the manufacturer and purchaser shall be used for determination of oxygen content.

4.3.2 Mechanical properties. Tensile test specimens shall be prepared and tested as specified in Method 211.1 (round specimens) of Federal Test Method Std. No. 151. Yield point shall be determined by the .2 percent offset method. Charpy V-notch impact specimens shall be prepared and tested as specified in Method 221.1 of Federal Test Method Std. No. 151. Testing shall be done on a government qualified test machine.

4.3.3 Density. The parts and test specimens shall be tested in accordance with ASTM Standard Test Method B-328.

4.3.4 Surface decarburization. A specimen for microscopic examination shall be taken from a test slice of 1/4 to 1/2 inch thickness and shall extend from the surface of the test slice to a depth greater than the allowable decarburization specified in 3.4. The specimen shall be polished and suitably etched. The depth of decarburization shall be determined using a magnification of 100X or greater.

4.3.5 Microstructure. A specimen shall be cut from each fractured Charpy V-notch impact specimen such that the surface to be examined shall be parallel to the fracture face and approximately 0.050 inch away. The specimen shall be suitably polished, etched, viewed at 100X, and compared to Figure 1 as specified in 3.5. Specimen location as taken from the production part shall be optional. However, the specimen shall be prepared as specified above and compared to Figure 1.

4.4 Rejection and retest. Failure of samples or specimens to comply with specified requirements shall be cause for rejection of the parts represented by the specimens or samples. However, at the discretion of the contractor, retesting will be permitted. Retest samples or specimens shall be tested to replace each failed sample or specimen. If one retest sample or specimen fails, the parts shall be rejected and additional retesting shall not be permitted.

PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking of parts shall be as specified in the applicable packaging document or in the contract.

6.1 This specification provides basic requirements for forged metal powder parts. Dimensional requirements shall be as specified on the applicable drawing or in the contract.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Applicable drawing or dimensional data.
c. Hardness level if required (see Table II).
d. Preparation for delivery requirements.

Custodians:
Army - WC
Navy - YD
Air Force - 11

Preparing activity:
Army - WC

Project No. FORG-0038

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-F-45961, Forgings, Prealloyed Steel Powder

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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